

Section: CS

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Multi-Signature Authentication Web Proxy

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To see what has changed since the last report, view the diff [here](#).

Problem Statement

Background and statement of the theoretical or practical problem, issue or opportunity motivating the research.

Enterprise grade cryptocurrency wallets have long implemented multi-signature (“multi-sig”) authentication, which requires two or more parties to sign an outbound transaction from a secure wallet before it is sent to the receiving party. This security mechanism ensures that before a transaction is sent, it is agreed upon by the group, rather than any one individual. But what if you want to require multiple parties to approve of a critical setting change or release a package? No existing general-purpose web authentication system (think Duo or Authelia) requires multiple distinct people to cooperate before granting access.

My practicum project proposes a general-purpose multi-signature authentication web proxy that allows access to the requested resource only when multiple parties ~~provide their secret keys~~authenticate and approve the request.

Supply chain attacks have been wreaking havoc on developers in recent months. Current solutions focus on locking down individual developer accounts, but I believe a better solution is requiring multiple human verifiers to approve a project release before it goes public. My project will provide a solution that, while generalizable, will be directly applicable to this use case. This project will demonstrate this gap and then advocate for platform-native functionality to be added to the repositories.

Solution Statement

This research will (the overall aim of the research) in order to (how the research will address the problem) with specific work product/deliverables described.

This research will create a general-purpose multi-signature authentication web proxy that, as an example of a real-world use case, will prevent any one individual from publishing a package to PyPI and instead require multiple reviewers to approve of the package’s release. I will develop an application that will be hosted on a web server. When a developer attempts to publish a package to PyPI, the web proxy will:

1. Intercept the request while saving it for publishing
2. Notify other maintainers that a package is attempting to be published
3. Gather enough approvals from the maintainers until a quorum is reached
4. Then forward the publish request on to the PyPI repository

Completed Tasks (Last 2 Weeks)

This Progress Report, I focused on front-loading the research, specifications, and architecture that the MVP of my project will need before committing to writing any code. I find this is ideal when working with AI to code as any knowledge it needs to create the product is all in one place. Using the /grill-me skill, I had long conversations using Claude code to fully develop the different systems within my project, memorialized in the /docs directory as specifications, Architecture Decision Records (ADRs), a Product Requirements Document (PRD), constraints the project must work in, and a threat model. These two weeks' efforts represent roughly 5,800 lines of specification created over 32 commits.

- ~~I brainstormed project ideas and created the initial Project Proposal~~Conducted broad cryptography research to ground the authentication scheme.
- ~~I continued to define the projects scope, motivation, use cases, and deliverables and incorporated them into the first progress report~~Evaluated and rejected multi-signature cryptography schemes.
 1. Ultimately, through research I decided to forgo complex and unnecessary multi-signature authentication cryptography solutions and instead decided on a standard credential-backed scheme for multiple reasons: (1) Multi-signature authentication schemes often require communication between users, (2) it would require every user to create and manage cryptographic keys, which is not ideal for non-technical users, and (3) it is overly complex when some of the security of the system would still rely on the security of the proxy server.
- Decided on and documented the authentication and cryptographic scheme.
- ~~I conducted research into real-world use cases for this project to better define the project's motivation~~Defined the system architecture and request data flow.
- ~~I defined a use case, publishing packages through PyPi, with a real world data flow.~~<https://github.com/Ian-Costa18/Cybersecurity-Practicum/blob/progress-report-2/Research/PyPiWr> application specifications for account management, approver authentication, configuration, notification system, and the web proxy.
- ~~I utilized AI to create a broad literature review on topics discussed in my project proposal for future reference. This broad literature review research AI-assisted report can be found here.~~ Scoped the MVP and assessed the system's security posture through a threat model.
- ~~I began targeted research into multi-signature authentication cryptography primitives.~~ Wrote a Product Requirements Document.
- Documented project decisions into ADRs.
- Ran a cross-document consistency audit and remediated 21 findings.
- Set up project process and the foundation for AI-assisted development.

Tasks for the Next Project Report

What will you do over the next two weeks for your project?

1. Decide on ~~a multi-sig authentication scheme.~~ the tech stack for implementation of the system.
2. ~~Secrets range from username/password combinations to tokens, how will these signatures be stored, and how will they be able to be used in a multi-signature scheme?~~ Write documentation to guide agents to the documents needed to create/edit parts of the system.
3. ~~Must be based on research peer-reviewed white papers~~ Break the implementation down into vertical implementation tasks.
4. ~~Create in-depth, researched application specifications-~~
 - ~~Flowchart from client machine F&E web proxy F&E notification to approvers F&E approvals F&E access allowed/denied to secure endpoint-~~
 - ~~Authentication systemspecification using multi-signature authentication scheme-~~
 - ~~Web proxy application specification-~~
 - ~~Approver notification system specification-~~
 - ~~Approval system specification-~~

Begin writing code to implement the MVP.

Questions I Have or Issues I'm Running Into

Are there unanticipated obstacles, is something taking longer than you thought it might, or is there an area where the Instructional Team might be able to provide some direction or clarification?

~~When researching use cases, I found it difficult to conceptualize how I'm going to integrate~~ I made many big decisions this Progress Report cycle that will ultimately shape how my project will come together. One example of this is the decision to use a more standard credential-backed authentication scheme rather than relying on the multi-signature authentication ~~into already existing platforms. For example, for the PyPI use case, the existing authentication system in PyPI is that you have an owner and multiple maintainers. The owner and the maintainers can all push packages to the public repository any time they wish. There is no way to restrict this functionality. This becomes a problem for my solution because I need to have fine-grained controls over who can push to the public repository. The solution I found is a bit of a hack. For the Web Proxy to be the only one with access to publishing rights on the repository, there will need to be a service account specific to the user's project that is the owner of the repository and there needs to be no maintainers. Then my web proxy will be the only one with the publishing token. Ideally, the solution is for every repository to have this multi-signature function. Therefore, my research and deliverables serve more cryptography schemes that I originally got the idea for this project from. However, I did not want to implement these schemes blindly just because I had sunk time and effort into~~

researching them. I believe that a credential-based solution is ultimately simpler for the users, simpler to implement for an MVP, and just as secure as a multi-signature-based scheme. I do think that it has the potential for implementation in future versions of this project, but as a proof-of-concept ~~and is instead built to show that this functionality is possible and should be a first-class authentication mechanism in package repositories. By making my project general use, this advocacy for multi-sig authentication functionality could be expanded to other use cases which I have not yet identified~~to champion the idea of requiring multiple users to approve a request before it goes through, this is the solution that I want to implement. This does represent a big pivot in my thought process for designing this project, but I believe it is for the best. I would welcome the instructional team's perspective on whether taking this step forward weakens to my project's core contribution in any way, or whether the multi-party-approval concept stands on its own as a proof-of-concept.

Methodology Paragraph Summary

Description of the methodology you are proposing to get from the starting point of your project to the final deliverable. Projects should employ some systematic process to get to your solution and a clearly identified way to evaluate it to ensure it solves the problem statement identified.

The project will follow an iterative and research-driven development process. I will start by conducting background research on multi-signature authentication cryptography primitives, existing authentication systems, and any other relevant security design patterns. That research will then inform application specifications for the proxy and approval flow. I will then build a minimum viable product that implements the core flow of intercepting a request, collecting approvals until a quorum is reached, and forwarding the request to the target endpoint. After reviewing the MVP, I will identify any missing requirements, limitations, and security concerns, then return to do additional research and refinement of the specification before implementing the full system. I will then develop an evaluation system that ensures my project solves the intended use cases I've identified. This evaluation system will likely consist of integration tests that verify that the project works for its intended use cases (usability), blocks attacks that attempt to bypass authentication mechanisms (security), and quantifies proxy overhead (performance), though optimizations are out of scope for this proof-of-concept.

Timeline

Enter one task per row. Include only project tasks (not peer feedback or progress report submissions). Tasks must be actionable. Update the Status column each week (Completed / In Progress / Cancelled / etc.).

Week	Description of Task	Status
W1	Brainstorm ideas for the project	Complete
W1	Create Project Proposal	Complete
W2	Conduct broad project research	Complete
W2	Refine Problem/Solution Statement and Methodology	Complete
W2	Define Project Timeline	Complete
W2	Conduct research on multi-auth crypto primitives	In Progress Complete
W3	Develop authentication scheme	Not Started Complete
W3	Define high-level architecture and data flow	Not Started Complete
W4	Write detailed application specifications	Not Started Complete
W4	Create harness for AI assisted programming	In Progress
W5	Decide on technology stack	Not Started
W5	Develop minimum viable product	Not Started
W6	Review MVP to identify missing requirements, limitations, and security concerns	Not Started
W6	Refine application specifications based on MVP review	Not Started
W7	Develop MVP evaluation framework based on use cases, security concerns, and performance	Not Started
W8	Begin building fully featured application based on project specifications	Not Started
W8	Create report outline	Not Started
W9	Continue iterating on application development	Not Started
W9	Begin writing report	Not Started
W10	Evaluate project based on security, usability, and performance	Not Started
W10	Continue writing report	Not Started
W11	Add finishing touches to project implementation	Not Started
W11	Finalize report	Not Started

Evaluation

Include any evaluation plans and/or results by Progress Report 3. This may expand as you finalize the report.

Report Outline

Include an outline of your final report by Progress Report 4. This may expand as you finalize the report.

References

List sources you have reviewed for your project.

Appendix

If there are notes, sources, or figures you want to reference, include them here. Draft work product should begin by Progress Report 3.